



SYMBIOSIS INSTITUTE OF TECHNOLOGY (SIT)

Constituent of Symbiosis International (Deemed University), Pune

(Established under Section 3 of the UGC Act of 1956 vide notification number F-9-12/2001-U-3 of the Government of India)
Re-Accredited by NAAC with 'A++' Grade

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Class : CSE
Batch : 2025-29

Assignment : 24

Subject: Programming with Java

ASSIGNMENT No : 24

TITLE :Develop a Java GUI application integrated with JDBC to perform database operations."

Problem Statement

Develop a Java GUI application integrated with JDBC to perform database operations."

Learning Objectives

To integrate Java GUI applications with databases using JDBC.

Theory

Java Swing provides graphical user interface components such as buttons, labels, text fields, tables, and forms, while JDBC enables database connectivity. Combining Swing with JDBC allows developers to create interactive desktop applications capable of performing database operations through a graphical interface. Users can enter information using GUI components, which is then processed and stored in the database using SQL queries. Retrieved records can be displayed in tables or text fields for easy viewing and modification. This integration improves user experience by eliminating command-line interaction and forms the foundation of desktop applications such as inventory management systems, library management systems, and student management systems.



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HTML File (index.html) : (GUI only , integration performed in previous experiments)

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Hospital Management System - Portals</title>
  <style>
    * {
      box-sizing: border-box;
      font-family: 'Segoe UI', Tahoma, Geneva, Verdana, sans-serif;
      margin: 0;
      padding: 0;
    }

    body {
      background-color: #eef2f5;
      display: flex;
      justify-content: center;
      align-items: center;
      height: 100vh;
    }

    .container {
      background-color: #ffffff;
      width: 400px;
      padding: 30px;
      border-radius: 8px;
      box-shadow: 0 4px 15px rgba(0, 0, 0, 0.1);
    }

    .tabs {
      display: flex;
      margin-bottom: 25px;
      border-bottom: 2px solid #e0e0e0;
    }

    .tab-btn {
      flex: 1;
```



```
padding: 10px;
text-align: center;
background: none;
border: none;
font-size: 15px;
font-weight: bold;
color: #777;
cursor: pointer;
outline: none;
}

.tab-btn.active {
  color: #007bff;
  border-bottom: 3px solid #007bff;
}

.form-group {
  margin-bottom: 20px;
}

.form-group label {
  display: block;
  font-size: 14px;
  color: #333;
  margin-bottom: 6px;
}

.form-group input {
  width: 100%;
  padding: 10px;
  border: 1px solid #ccc;
  border-radius: 4px;
  font-size: 14px;
}

.btn {
  width: 100%;
  padding: 12px;
  background-color: #007bff;
  color: white;
```



```
border: none;
border-radius: 4px;
font-size: 16px;
font-weight: bold;
cursor: pointer;
transition: background 0.3s;
}

.btn:hover {
    background-color: #0056b3;
}

.form-section {
    display: none;
}

.form-section.active {
    display: block;
}

.status-msg {
    margin-top: 15px;
    font-size: 14px;
    text-align: center;
    font-weight: bold;
}
</style>
</head>
<body>

<div class="container">
    <div class="tabs">
        <button class="tab-btn active" onclick="switchTab('user-login', this)">General
Login</button>
        <button class="tab-btn" onclick="switchTab('staff-login', this)">Staff Login</button>
    </div>

    <!-- General User Login Form -->
    <div id="user-login" class="form-section active">
        <h2 style="text-align: center; margin-bottom: 20px; color: #333;">System Login</h2>
```



```
<div class="form-group">
  <label for="username">Username</label>
  <input type="text" id="username" placeholder="Enter Username">
</div>
<div class="form-group">
  <label for="user-password">Password</label>
  <input type="password" id="user-password" placeholder="Enter Password">
</div>
<button class="btn" onclick="handleUserLogin()">Login</button>
<div id="user-msg" class="status-msg"></div>
</div>

<!-- Hospital Staff Login Form -->
<div id="staff-login" class="form-section">
  <h2 style="text-align: center; margin-bottom: 20px; color: #333;">Hospital Staff
Portal</h2>
  <div class="form-group">
    <label for="staff-id">Staff ID</label>
    <input type="text" id="staff-id" placeholder="e.g., DOC101 or NUR201">
  </div>
  <div class="form-group">
    <label for="staff-password">Password</label>
    <input type="password" id="staff-password" placeholder="Enter Password">
  </div>
  <button class="btn" onclick="handleStaffLogin()">Login to Portal</button>
  <div id="staff-msg" class="status-msg"></div>
</div>
</div>

<script>
function switchTab(tabId, element) {
  document.querySelectorAll('.form-section').forEach(el => el.classList.remove('active'));
  document.querySelectorAll('.tab-btn').forEach(el => el.classList.remove('active'));
  document.getElementById(tabId).classList.add('active');
  element.classList.add('active');
}

function handleUserLogin() {
  const user = document.getElementById('username').value;
  const pass = document.getElementById('user-password').value;
```



```
const msg = document.getElementById('user-msg');

if (user === "admin" && pass === "admin123") {
    msg.style.color = "green";
    msg.innerText = "Login Successful! Welcome, " + user + ".";
} else {
    msg.style.color = "red";
    msg.innerText = "Invalid Username or Password. Access Denied.";
}
}

function handleStaffLogin() {
    const id = document.getElementById('staff-id').value;
    const pass = document.getElementById('staff-password').value;
    const msg = document.getElementById('staff-msg');

    if (id === "DOC101" && pass === "docpass") {
        msg.style.color = "green";
        msg.innerText = "Access Granted: Welcome, Dr. " + id + ". Redirecting to Doctor
Dashboard.";
    } else if (id === "NUR201" && pass === "nurpass") {
        msg.style.color = "green";
        msg.innerText = "Access Granted: Welcome, Nurse " + id + ". Redirecting to Nursing
Station.";
    } else {
        msg.style.color = "red";
        msg.innerText = "Authentication Failed: Invalid Staff ID or Password.";
    }
}
}
</script>

</body>
</html>
```



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Output : Case 1

[General Login](#) [Staff Login](#)

System Login

Username

Password

Login

Login Successfull! Welcome, admin.

Case 2

[General Login](#) [Staff Login](#)

System Login

Username

Password

Login

Invalid Username or Password. Access Denied.



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Case 3 :

[General Login](#) [Staff Login](#)

Hospital Staff Portal

Staff ID

Password

[Login to Portal](#)

Access Granted: Welcome, Dr. DOC101. Redirecting to Doctor Dashboard.

Case 4

[General Login](#) [Staff Login](#)

Hospital Staff Portal

Staff ID

Password

[Login to Portal](#)

Authentication Failed: Invalid Staff ID or Password.



Conclusion : The Hospital Management System GUI was successfully implemented using HTML, CSS, and JavaScript. The GUI provides separate General Login and Hospital Staff Login sections with a clean and user-friendly interface. JavaScript was used to handle tab switching, user input, authentication, and display of login status messages. The implementation demonstrates how a functional login-based GUI can be designed for a hospital management system.

Exercise

Q1) Create a Library Management System GUI connected with database

Sol) Code :

```
import javax.swing.*;
import java.awt.*;
import java.sql.*;

public class LibraryManagement extends JFrame {

    private static final String DB_URL = "jdbc:mysql://localhost:3306/hospital_db";
    private static final String DB_USER = "root";
    private static final String DB_PASSWORD = "Abcd@1234";

    JTextField bookName, bookId, studentName, issueDate, returnDate;
    JTextArea output;

    public LibraryManagement() {
        setTitle("Library Management System");
        setSize(600, 500);
        setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
        setLocationRelativeTo(null);

        JTabbedPane tabs = new JTabbedPane();

        JPanel bookPanel = new JPanel(new GridLayout(4, 2, 10, 10));

        bookName = new JTextField();

        JButton addBook = new JButton("Add Book");
        JButton viewBooks = new JButton("View Books");
```



```
bookPanel.add(new JLabel("Book Name:"));
bookPanel.add(bookName);
bookPanel.add(addBook);
bookPanel.add(viewBooks);

JPanel issuePanel = new JPanel(new GridLayout(5, 2, 10, 10));

bookId = new JTextField();
studentName = new JTextField();
issueDate = new JTextField();
returnDate = new JTextField();

JButton issueBook = new JButton("Issue Book");
JButton viewIssued = new JButton("View Issued Books");

issuePanel.add(new JLabel("Book ID:"));
issuePanel.add(bookId);

issuePanel.add(new JLabel("Student Name:"));
issuePanel.add(studentName);

issuePanel.add(new JLabel("Issue Date (YYYY-MM-DD):"));
issuePanel.add(issueDate);

issuePanel.add(new JLabel("Return Date (YYYY-MM-DD):"));
issuePanel.add(returnDate);

issuePanel.add(issueBook);
issuePanel.add(viewIssued);

output = new JTextArea();
output.setEditable(false);

tabs.addTab("Books", bookPanel);
tabs.addTab("Issue Book", issuePanel);

add(tabs, BorderLayout.NORTH);
add(new JScrollPane(output), BorderLayout.CENTER);
```



```
addBook.addActionListener(e -> addBook());
viewBooks.addActionListener(e -> viewBooks());
issueBook.addActionListener(e -> issueBook());
viewIssued.addActionListener(e -> viewIssuedBooks());
}

private Connection getConnection() throws SQLException {
    return DriverManager.getConnection(DB_URL, DB_USER, DB_PASSWORD);
}

private void addBook() {
    String name = bookName.getText();

    String query = "INSERT INTO books (name) VALUES (?)";

    try (Connection conn = getConnection();
        PreparedStatement pstmt = conn.prepareStatement(query)) {

        pstmt.setString(1, name);
        pstmt.executeUpdate();

        output.setText("Book added successfully.");
        bookName.setText("");

    } catch (Exception e) {
        output.setText("Database Error: " + e.getMessage());
    }
}

private void viewBooks() {
    String query = "SELECT * FROM books";

    try (Connection conn = getConnection();
        Statement stmt = conn.createStatement();
        ResultSet rs = stmt.executeQuery(query)) {

        output.setText("BOOKS\n\n");

        while (rs.next()) {
```



```
        output.append(
            "ID: " + rs.getInt("id") +
            " Name: " + rs.getString("name") +
            "\n"
        );
    }

} catch (Exception e) {
    output.setText("Database Error: " + e.getMessage());
}

}

private void issueBook() {
    String query = "INSERT INTO book_issue " +
        "(book_id, student_name, issue_date, return_date) " +
        "VALUES (?, ?, ?, ?)";

    try (Connection conn = getConnection();
        PreparedStatement pstmt = conn.prepareStatement(query)) {

        pstmt.setInt(1, Integer.parseInt(bookId.getText()));
        pstmt.setString(2, studentName.getText());
        pstmt.setDate(3, Date.valueOf(issueDate.getText()));
        pstmt.setDate(4, Date.valueOf(returnDate.getText()));

        pstmt.executeUpdate();

        output.setText("Book issued successfully.");

        bookId.setText("");
        studentName.setText("");
        issueDate.setText("");
        returnDate.setText("");

    } catch (Exception e) {
        output.setText("Database Error: " + e.getMessage());
    }

}
```



```
private void viewIssuedBooks() {
    String query = "SELECT * FROM book_issue";

    try (Connection conn = getConnection();
        Statement stmt = conn.createStatement();
        ResultSet rs = stmt.executeQuery(query)) {

        output.setText("ISSUED BOOKS\n\n");

        while (rs.next()) {
            output.append(
                "Book ID: " + rs.getInt("book_id") +
                " Student: " + rs.getString("student_name") +
                " Issue Date: " + rs.getDate("issue_date") +
                " Return Date: " + rs.getDate("return_date") +
                "\n"
            );
        }

    } catch (Exception e) {
        output.setText("Database Error: " + e.getMessage());
    }
}

public static void main(String[] args) {
    SwingUtilities.invokeLater(() -> {
        new LibraryManagement().setVisible(true);
    });
}
```



SQL Table :

```
1 • CREATE DATABASE hospital_db;
2 • USE hospital_db;
3 • USE hospital_db;
4
5 • CREATE TABLE books (
6     id INT AUTO_INCREMENT PRIMARY KEY,
7     name VARCHAR(100)
8 );
9
10 • CREATE TABLE book_issue (
11     book_id INT,
12     student_name VARCHAR(100),
13     issue_date DATE,
14     return_date DATE
15 );
16
```

Output : Add books

Library Management System

Books Issue Book

Book Name:

Alchemist

Add Book

View Books

Book added successfully



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View Books :

Library Management System

Books Issue Book

Book Name:

Add Book

View Books

BOOKS

ID: 1 Name: Harry Potter
ID: 2 Name: Alchemist
ID: 3 Name: Mein Kampf
ID: 4 Name: Metamorphosis
ID: 5 Name: Immortals of Meluha

Book Issue:

Library Management System

Books Issue Book

Book ID:

Student Name:

Issue Date (YYYY-MM-DD):

Return Date (YYYY-MM-DD):

Issue Book View Issued Books

Book issued successfully.



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View Issued Books :

Library Management System

Books Issue Book

Book ID:

Student Name:

Issue Date (YYYY-MM-DD):

Return Date (YYYY-MM-DD):

Issue Book View Issued Books

ISSUED BOOKS

Book ID: 2 Student: Atharva Issue Date: 2026-08-30 Return Date: 2026-09-15

Book ID: 3 Student: Vatsal Issue Date: 2026-09-01 Return Date: 2026-09-23

Q2) Create a Book Issue Tracking System GUI application using Java Swing connected to a database. The system should store book ID, student name, issue date, and return date, and allow record management.

Sol) Code :

```
import javax.swing.*;
import java.sql.*;
```

```
public class BookIssue {
    public static void main(String[] args) {
        JFrame f = new JFrame("Book Issue Tracking");

        JLabel l1 = new JLabel("Book ID:");
        JLabel l2 = new JLabel("Student Name:");
        JLabel l3 = new JLabel("Issue Date:");
        JLabel l4 = new JLabel("Return Date:");

        JTextField bookId = new JTextField();
```



```
JTextField student = new JTextField();
JTextField issue = new JTextField();
JTextField ret = new JTextField();

JButton add = new JButton("Issue Book");
JButton view = new JButton("View Records");

l1.setBounds(30, 30, 100, 25);
bookId.setBounds(130, 30, 200, 25);

l2.setBounds(30, 70, 100, 25);
student.setBounds(130, 70, 200, 25);

l3.setBounds(30, 110, 100, 25);
issue.setBounds(130, 110, 200, 25);

l4.setBounds(30, 150, 100, 25);
ret.setBounds(130, 150, 200, 25);

add.setBounds(60, 200, 120, 30);
view.setBounds(190, 200, 130, 30);

f.add(l1);
f.add(bookId);
f.add(l2);
f.add(student);
f.add(l3);
f.add(issue);
f.add(l4);
f.add(ret);
f.add(add);
f.add(view);

add.addActionListener(e -> {
    try {
        Connection con = DriverManager.getConnection(
            "jdbc:mysql://localhost:3306/hospital_db",
            "root", "Abcd@1234");
```



```
PreparedStatement ps = con.prepareStatement(
    "INSERT INTO book_issue VALUES(?,?,?,?)");

ps.setInt(1, Integer.parseInt(bookId.getText()));
ps.setString(2, student.getText());
ps.setDate(3, Date.valueOf(issue.getText()));
ps.setDate(4, Date.valueOf(ret.getText()));

ps.executeUpdate();

JOptionPane.showMessageDialog(f, "Book Issued Successfully");
con.close();

} catch (Exception ex) {
    JOptionPane.showMessageDialog(f, "Error");
}
});

view.addActionListener(e -> {
    try {
        Connection con = DriverManager.getConnection(
            "jdbc:mysql://localhost:3306/hospital_db",
            "root", "Abcd@1234");

        Statement st = con.createStatement();
        ResultSet rs = st.executeQuery("SELECT * FROM book_issue");

        String data = "";

        while (rs.next()) {
            data += "Book ID: " + rs.getInt(1)
                + " | Student: " + rs.getString(2)
                + " | Issue: " + rs.getDate(3)
                + " | Return: " + rs.getDate(4) + "\n";
        }

        JOptionPane.showMessageDialog(f, data);
        con.close();
    }
});
```



```
    } catch (Exception ex) {  
        JOptionPane.showMessageDialog(f, "Error");  
    }  
});  
  
f.setSize(380, 290);  
f.setLayout(null);  
f.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);  
f.setLocationRelativeTo(null);  
f.setVisible(true);  
}  
}
```

SQL Table Used :

```
10 • CREATE TABLE book_issue (  
11     book_id INT,  
12     student_name VARCHAR(100),  
13     issue_date DATE,  
14     return_date DATE  
15 );
```



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Output : Issue Book :

Book Issue Tracking

Book ID: 1

Student Name: Avdhoot

Issue Date: 2026-08-15

Return Date: 2026-09-12

Issue Book View Records

Message

Book Issued Successfully

OK

View Issued Books : (Records)

Book Issue Tracking

Book ID: 1

Student Name: Avdhoot

Issue Date: 2026-08-15

Return Date: 2026-09-12

Issue Book View Records

Message

Book ID: 2 | Student: Atharva | Issue: 2026-08-30 | Return: 2026-09-15
Book ID: 3 | Student: Vatsal | Issue: 2026-09-01 | Return: 2026-09-23
Book ID: 1 | Student: Avdhoot | Issue: 2026-08-15 | Return: 2026-09-12

OK